EXPERIMENT - 01

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Analysis, Development and Verification of 8-bit and 16-bit Arithmetic Operations Using Assembly Language Program

a) Develop and Verification of (i) 8-Bit Addition and (ii) Subtraction Using 8-Bit Microprocessor Assembly Language Programming

b) Verification of (iii)16-Bit Mul and (iv) Division Using 8-Bit Microprocessor Assembly Language Programming

OBJECTIVE:

To create, implement, and test assembly language programs to perform 8-bit addition and subtraction and 16-bit multiplication and division using 8-bit microprocessor.

ALGORITHM:

- Prepare the data and registers for arithmetic operations.
- Input the required data for arithmetic operations into registers or memory.
- Calculate the required arithmetic operation using assembly language instructions.
- Store the results into destination memory or register.
- Check the results by comparing them with expected results.

(a) (i) 8-BIT ADDITION

Formula

$$\text{Sum} = A + B$$

Input

- First number: 25H
- Second number: 15H

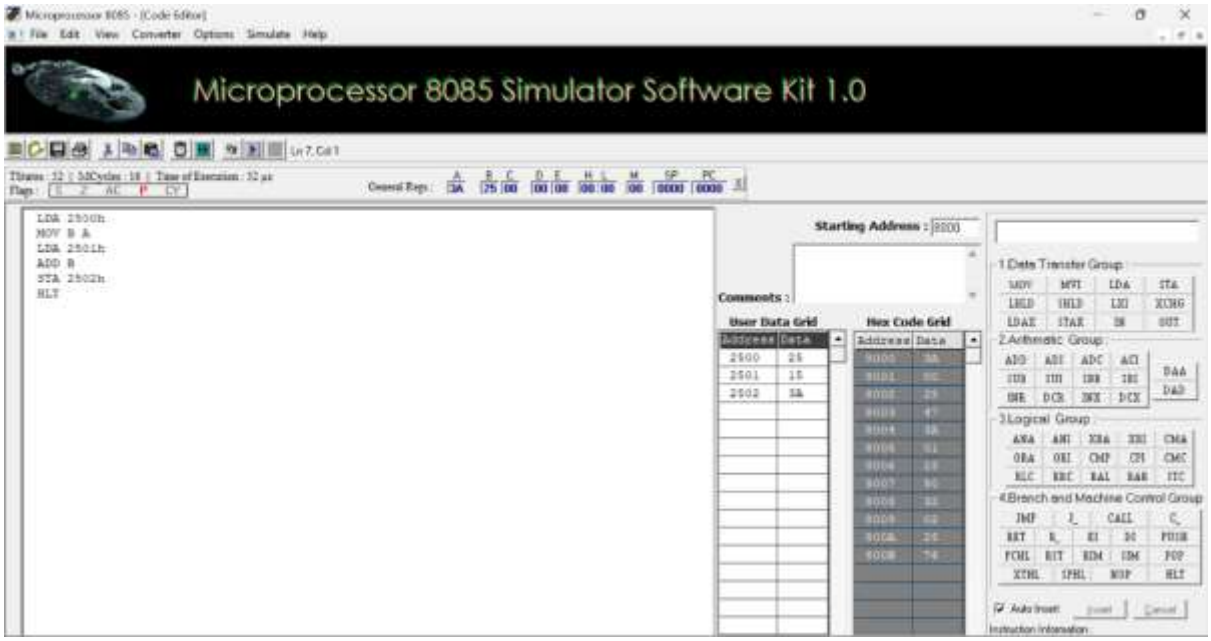
Code (8085 Assembly Language)

```
LXI H, 2000H
MOV A, M
INX H
ADD M
INX H
MOV M, A
HLT
```

Output

- Memory location 2000H = 25H
- Memory location 2001H = 15H
- Memory location 2002H = 3AH
- Result: 25H + 15H = 3AH

EXECUTION:



(a) (ii) 8-BIT SUBTRACTION

Formula

$$\text{Difference} = A - B$$

Input

- Minuend: 35H
- Subtrahend: 12H

Code (8085 Assembly Language)

LXI H, 2000H

MOV A, M

INX H

SUB M

INX H

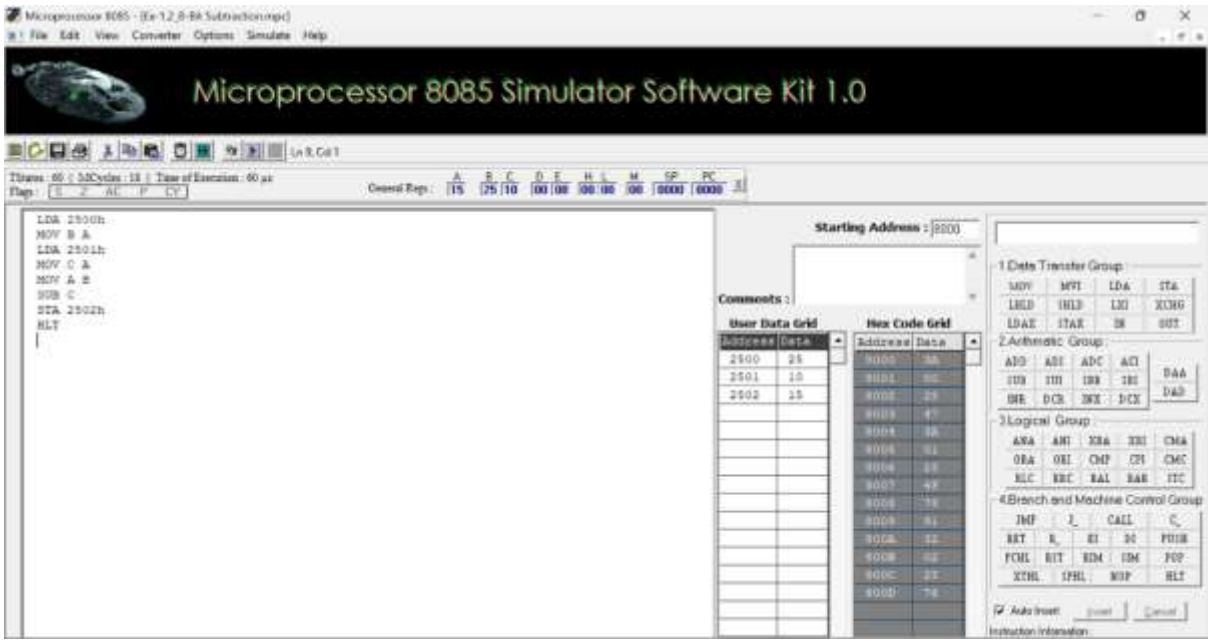
MOV M, A

HLT

Output

- Memory location 2000H = 35H
- Memory location 2001H = 12H
- Memory location 2002H = 23H
- Result: 35H − 12H = 23H

EXECUTION:



(b) (iii) 16-BIT MULTIPLICATION

Formula

$$\text{Product} = A \times B$$

Input

- Multiplicand: 0012H (18 decimal)
- Multiplier: 0003H (3 decimal)

Code (8085 Assembly Language)

```
LXI H, 2000H
MOV B, M
INX H
MOV C, M
LXI H, 0000H
LOOP: MOV A, L
ADD B
MOV L, A
JNC NEXT
INR H
NEXT: DCR C
JNZ LOOP
SHLD 2002H
HLT
```

Output

- Memory location 2000H = 12H
- Memory location 2001H = 03H
- Memory location 2002H = 36H (Low byte)
- Memory location 2003H = 00H (High byte)
- Result: 12H × 03H = 0036H

EXECUTION:



(b) (iv) 16-BIT DIVISION

Note: The 8085 microprocessor does not have a direct division instruction. Repeated subtraction is used to divide a 16-bit dividend by an 8-bit divisor and obtain the quotient and remainder.

Formula

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

Input

- Dividend: 001EH (30 decimal)
- Divisor: 03H (3 decimal)

Code (8085 Assembly Language)

LXI H, 2000H

MOV A, M

INX H

MOV B, M

MVI C, 00H

LOOP: CMP B

JC DONE

SUB B

INR C

JMP LOOP

DONE: STA 2002H

MOV A, C

STA 2003H

HLT

Output

- Memory location 2000H = 1EH (Dividend)
- Memory location 2001H = 03H (Divisor)
- Memory location 2002H = 00H (Remainder)
- Memory location 2003H = 0AH (Quotient)
- Result: 30 ÷ 3 = 10, Remainder = 0

EXECUTION:



STUDENT REFLECTION:

The experiment taught me how to perform arithmetical processes with the help of assembly programming. The experiment involved 8-bit addition and subtraction and 16-bit calculating of results and confirming them.

RESULT

Thus, the assembly language programs for 8-bit addition, 8-bit subtraction, 16-bit multiplication, and division were developed and verified successfully using the 8085 microprocessor instruction set.